

MADHAV KUMAR

Software Engineer | Backend & Cloud

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PROFILE

Software Engineering intern at Nielsen with hands-on experience in building and optimizing large-scale distributed systems, improving system reliability, and automating production workflows using AWS, Python, and Java. Strong in backend development, cloud infrastructure, and CI/CD-driven deployments.

TECHNICAL SKILLS

- **Languages**
 - Python, Java
- **Cloud and Infrastructure**
 - AWS (Lambda, S3, EC2, RDS, SQS, SNS, CloudWatch, IAM), Docker, Kubernetes, Terraform
- **Databases**
 - PostgreSQL, ClickHouse, MySQL
- **Backend and Tools**
 - REST APIs, Microservices, Linux, GitLab CI/CD, Git, Jira, Confluence

EXPERIENCE

Nielsen - Software Development Engineer Intern

Mar 2025 – Present

Audio Quality Engine

- Developed a Python tool that automatically detects audio problems like noise, muffled sound, and distortion, flagging clips that fall below acceptable quality before they reach the next stage of the pipeline.
- Resolved critical audio capture failures affecting 2,000 plus daily pipeline runs, cutting dropout rates from 22% to under 1% and restoring full SLA compliance.
- Stabilized capture pipelines by tuning audio buffer settings and standardizing configurations end to end, eliminating signal loss that was breaking recordings mid capture.

Infrastructure and Automation

- Introduced WAV to FLAC compression across media pipelines, cutting storage usage by 70%, and set up event tracking using PostgreSQL, SNS, SQS, and ClickHouse.
- Removed redundant execution patterns from existing workflows, improving pipeline speed by 34%.
- Rolled out configuration changes across 450 plus Windows VMs in multiple regions using PowerShell scripts, keeping all machines consistent without manual effort.
- Designed an AWS Lambda validation tool integrated with CloudWatch that cut manual log review by 60% and added automated checks into CI/CD release flows.

PROJECTS

- **GitForgeX - Version Control System** - Python, Docker, REST APIs
 - Developed a Git-like version control system from scratch using separate containerized services for managing repositories and tracking version history.
 - Each service communicates through APIs with a secure handshake for push operations and snapshot based storage so reads stay fast even as history grows.
- **Kubernetes Logging System** - Kubernetes, Fluentd, ClickHouse
 - Designed a logging setup where each running service automatically sends its logs to a central store with no manual collection needed.
 - Tested with 50,000 plus log events, all searchable and retrievable in under a second.

CERTIFICATES

Application Development in Java

- Gained practical knowledge in Java GUI development, JDBC for database management, and Java networking. Applied these skills in projects to improve application design and functionality.

EDUCATION

Lovely Professional University

Bachelor of Technology-Computer Science & Engineering - 87.90%

Jalandhar, Punjab, India

2022 - 2026